1. Perform exercise 4 at the end of Chapter 3 of the textbook.

Q1. This chapter discussed the role of values and beliefs in forming an organization's culture. The topic of organizational culture is big business on the Internet. Many companies use their Web pages to describe their mission, vision, and corporate values and beliefs. There also are many consulting firms that advertise how they help organizations change their culture. The purpose of this exercise is for you to obtain information pertaining to the organizational culture for two different companies. You can go about this task by very simply searching the key words organizational culture or corporate vision and values. This search will identify numerous companies for you to use to answer the following questions. You may want to select companies that you would like to work for in the future.

a. What are the espoused values and beliefs of the companies?

b. Use the worksheet in Figure 3.6 to assess the Web page. What does the Web page reveal about the culture of this organization? Would this culture be conducive to effective project management?

Ans.

a. The Driving Principles Behind NVIDIA's Success

NVIDIA's core values—collectively known as "Our Code"—serve as the foundation for how the company operates and innovates in a fast-paced, competitive tech industry. Each value plays a crucial role in shaping the company's culture and success:

• **Innovation:** At NVIDIA, innovation is more than just a buzzword—it's a call to continuously explore new ideas, experiment boldly, and push the boundaries of what technology can achieve. This relentless pursuit of creativity fuels breakthroughs in graphics processing, AI, and computing, enabling NVIDIA to stay ahead of the curve.

• **Excellence and Determination:** Excellence means setting and meeting the highest standards in every aspect of work. Determination complements this by embodying persistence—even when facing tough challenges or setbacks. Together, these values ensure that NVIDIA not only creates cutting-edge products but also delivers them with quality and reliability.

• **Speed and Agility:** The technology landscape changes rapidly, and NVIDIA's ability to move fast and adapt quickly to new trends or obstacles allows it to capitalize on emerging opportunities. This agility is key to maintaining leadership in innovation while staying responsive to customer needs and market demands.

• **Intellectual Honesty:** Transparency and truthfulness build trust internally and externally. By admitting mistakes openly and learning from them, NVIDIA cultivates a culture of continuous improvement and integrity, which strengthens teamwork and decision-making.

• **One Team:** Collaboration across departments and levels of the company fosters unity and shared purpose. This "One Team" mindset breaks down silos and encourages knowledge-sharing, enabling NVIDIA to combine diverse talents and perspectives into cohesive, high-impact solutions.

Together, these core values create a vibrant environment where employees feel empowered to innovate, excel, and collaborate—fuelling NVIDIA's ongoing success as a leader in technology.

b. Nvidia

1. Physical Characteristics:

NVIDIA's main office, or global headquarters, is located in Santa Clara, California, at 2788 San Tomas Expressway, right in the heart of Silicon Valley. This campus serves as the central hub for NVIDIA's research and development, engineering, and corporate functions. The headquarters consists primarily of two large buildings: Voyager, which spans 750,000 square feet, and Endeavor, covering 500,000 square feet. The design of these buildings features a futuristic style, incorporating triangular motifs that symbolize NVIDIA's roots in 3D graphics, where triangles play a key role in rendering. Sustainability is also a key focus in the architecture, with features such as ample skylights to bring in natural light and environmentally conscious building practices. In recent years, NVIDIA has significantly expanded its footprint in Santa Clara by acquiring several additional office complexes, including an eight-building facility totalling 550,000 square feet and a 500,000-square-foot campus connected to its main site. These expansions reflect the company's ongoing commitment to growth and innovation in Silicon Valley. For contact, the main phone number is +1 (408) 486-2000, and inquiries can be sent to info@nvidia.com.

2. Public Documents:

NVIDIA makes a wide range of public documents available to provide transparency into its corporate governance, financial performance, and operational policies. These documents can be accessed through the company's official website and regulatory filings. Key governance documents include the Code of Conduct, which sets ethical standards for all employees and directors, the Finance Team Code of Conduct that focuses on financial integrity, Global Tax Principles, corporate governance policies, and committee charters outlining the responsibilities of various board committees. NVIDIA also publishes detailed financial reports, such as annual reports that cover the company's performance and strategic outlook. As a publicly traded company, NVIDIA files regular reports with the U.S. Securities and Exchange Commission (SEC), including the annual Form 10-K, quarterly Form 10-Q, and current event Form 8-K filings. Additionally, NVIDIA maintains company policies addressing areas such as anti-corruption, human rights, environmental and safety practices, and privacy protection. All these documents are organized and available on NVIDIA's investor relations and company policy web pages, ensuring stakeholders have access to comprehensive and up-to-date information about the company's operations and governance.

3. Behaviour:

NVIDIA fosters a corporate culture grounded in integrity, respect, and innovation, which is clearly reflected in its Code of Conduct. The company expects all employees, from executives to new hires, to uphold the highest ethical standards in their daily work and interactions. Central to NVIDIA's behaviour is intellectual honesty—being transparent, admitting mistakes, and learning from them—which builds trust both within the company and with external partners. Employees are encouraged to act with fairness, respect diversity, and collaborate as "One Team," emphasizing unity and mutual support. The Code of Conduct also mandates compliance with all applicable laws and regulations, while reinforcing NVIDIA's commitment to social responsibility, including anti-corruption practices and respect for human rights. This code serves not only as a guideline for proper behaviour but also as a foundation for maintaining NVIDIA's reputation as a responsible and forward-thinking leader in technology.

4. Folklore:

NVIDIA was founded in 1993 by Jensen Huang, Chris Malachowsky, and Curtis Priem in Santa Clara, California. The company started with a vision to revolutionize graphics processing and computing. Its early folklore is closely tied to its breakthrough innovation in graphics processing units (GPUs), which transformed the gaming and professional visualization industries. NVIDIA's first major success came with the release of the RIVA series GPUs in the late 1990s, but the real game-changer was the introduction of the GeForce 256 in 1999, often called the world's first GPU, which set the standard for graphics acceleration. The company's culture and folklore emphasize innovation, perseverance, and a passion for solving difficult technical challenges.

Jensen Huang, NVIDIA's charismatic CEO, is often credited with inspiring a startup-like culture even as the company grew into a global tech giant. Stories of intense product development cycles, all-nighters, and bold bets on emerging technologies like AI and deep learning have become part of NVIDIA's lore.

Over time, NVIDIA evolved from being primarily a graphics card manufacturer to a leading force in AI computing, autonomous vehicles, and data canter solutions. This transformation is often seen as a testament to the company's willingness to innovate beyond its original focus and enter new frontiers. The folklore of NVIDIA also includes tales of its relentless drive for excellence and its collaborative "One Team" spirit, which has helped it navigate the highly competitive semiconductor industry.

We also have additional companies which we want to join such as Google and OpenAI because its vision of "organizing the world's information and making it universally accessible and useful" aligns with my passion for using data and technology to solve real-world problems. Google's values—innovation, integrity, respect for diversity, and focus on the user—create an environment where creativity and responsibility go hand in hand. Working in such a culture would allow me to grow professionally while contributing to projects that make a global impact. The company's emphasis on collaboration, inclusion, and sustainability also shows that Google is dedicated not just to technology, but to building a better future.

Google

I. Physical Characteristics:

Headquarters at the Googleplex in Mountain View, California. Open office layouts with colourful décor, beanbags, cafés, and outdoor spaces. Employees dress casually (jeans, T-shirts, hoodies). The design emphasizes creativity, collaboration, and comfort over hierarchy.

II. Public Documents:

"Our mission is to organize the world's information and make it universally accessible and useful." Core principles include: Focus on the user, act with integrity, innovate boldly, and respect diversity. Google also publishes AI Principles, sustainability goals (carbon-free energy by 2030), and annual diversity reports.

III. Behaviour:

Fast-paced and innovative, with brainstorming sessions, open discussions, and flat communication styles. Employees freely share ideas with managers. Collaboration and experimentation are encouraged, including the "20% time" policy for side projects. Work-life balance varies by team but flexibility is valued.

IV. Folklore:

Stories of Google starting in a Stanford dorm room with Larry Page and Sergey Brin. The launch of Gmail as a side project is legendary. Google Doodles represent its playful culture.

Tales of big bets like YouTube and Android acquisitions show its boldness. The “20% time” folklore inspires employees to dream big and take risks.

We also want to join OpenAI because its mission of ensuring that AGI benefits all of humanity aligns with my belief that technology should create positive, global impact. The company’s values—safety, transparency, collaboration, and technical leadership—match my passion for building AI responsibly while pushing the boundaries of innovation. Working in such a culture would give me the opportunity to grow alongside world-class researchers, contribute to ethical AI development, and be part of an organization shaping the future of technology for the greater good.

OpenAI

I. Physical Characteristics:

OpenAI’s main office is in San Francisco, California, located in the Mission District. The workspace is designed to be modern, minimalist, and collaborative, with open seating plans that encourage interaction among researchers and engineers. The décor reflects simplicity and focus, with whiteboards and brainstorming areas central to the office layout. Employees usually dress casually—jeans, T-shirts, or hoodies—reinforcing a relaxed yet focused work culture cantered on problem-solving rather than hierarchy.

II. Public Documents:

OpenAI’s mission is “to ensure that artificial general intelligence (AGI) benefits all of humanity.” Its published values highlight broadly shared benefit, long-term safety, technical leadership, and cooperative orientation. OpenAI frequently shares research papers, safety reports, and policy blogs to be transparent about its progress and challenges. The company has also released charters that commit it to prioritizing human welfare over profit and to avoiding harmful applications of AI.

III. Behaviour:

The pace at OpenAI is intense and research-driven, with teams working on ambitious, cutting-edge problems in artificial intelligence. Collaboration is highly emphasized, with interdisciplinary teams of engineers, researchers, and policy experts working closely together. Communication is open, and feedback is valued across all levels. Ethical discussions about AI safety, fairness, and alignment are a regular part of the work environment. While workloads can be demanding, the company also encourages creativity and careful reflection before deploying new technologies.

IV. Folklore:

OpenAI was founded in 2015 by a group including Elon Musk and Sam Altman, with the vision of making AI development open and safe. Early stories include its pledge to share research openly with the public and its bold move to transition into a “capped-profit” company to fund expensive AI research while sticking to its mission. The success of GPT-2 and GPT-3, and later ChatGPT, are often cited as milestones that changed public perception of AI. Tales of long nights of coding breakthroughs, debates on AI safety, and the

development of models that surprised even their creators form part of OpenAI's cultural lore.

2. Perform exercise 6 at the end of chapter 4 of the textbook. This is the one that focuses on deliverables and organizational units. Be clear about what the project is before addressing the other particulars of the exercise.

Q2. Use an example of a project you are familiar with or are interested in. Identify the deliverables and organization units (people) responsible. How would you code your system? Give an example of the work packages in one of your cost accounts.

Ans. Med-PaLM is a large language model developed by Google, adapted from the PaLM (Pathways Language Model) family and specialized for the medical field. It is designed to answer medical questions, summarize literature, and even work with images in its later versions. Med-PaLM has shown strong performance on medical benchmarks like USMLE-style questions, and it is being researched for safe, accurate, and trustworthy use in healthcare.

We chose this topic for our data science project because it is modern, impactful, and highly relevant. It combines data science with healthcare, which makes it meaningful and socially useful. It also gives us the opportunity to explore how AI can be applied to real-world, high-stakes problems where accuracy and safety matter. From a project management perspective, Med-PaLM involves clear tasks such as data preparation, model evaluation, visualization, and reporting, which makes it a strong example for applying project management principles. In addition, it allows us to practice working with cutting-edge AI, understand challenges like ethics and bias in medical data, and gain experience in organizing a complex interdisciplinary project.

Deliverables

1. Literature Review & Background – survey of Med-PaLM, Med-PaLM 2, and Med-PaLM research.

2. Dataset Preparation – collection, cleaning, de-identification, and splitting of medical datasets.

3. Model Setup & Fine-Tuning – environment setup, fine-tuning, hyperparameter tuning, and checkpointing.

4.Compute & Technical Specifications – Cloud compute setup, GPUs/TPUs, Storage (2–5 TB), RAM (≥128 GB), high-bandwidth networking

5.Evaluation & Safety Analysis – benchmark testing, clinical usefulness review, safety and bias checks.

6.Prototype Dashboard / Interface – simple UI for model queries, visualization of outputs and metrics.

7.Final Report & Presentation – APA-formatted report, slides, and demo.

Organizational Units (People Responsible)

1.Project Manager (PMO) – oversees planning, scheduling, and deliverables.

2.Data Engineer (DE) – handles dataset sourcing, cleaning, storage, and preprocessing.

3.Machine Learning Engineer (MLE) – manages model setup, fine-tuning, evaluation, and inference API.

4.Compute/Infrastructure Engineer (CIE) – sets up cloud environment, GPU/TPU clusters, storage, networking, and monitoring.

5.Domain Expert (Healthcare/Medical SME) – validates medical correctness and ensures ethical outputs.

6.Quality & Safety Officer (QA) – performs bias, safety, and harm reduction checks.

7.Visualization & UI Developer (PD) – builds prototype interface, dashboards, and prepares visuals/slides.

System Coding (Work Breakdown Structure):

- **1.0** Med-PaLM Project
  - **1.1** Research & Planning
    - **1.1.1** Literature review
    - **1.1.2** Define project scope
  - **1.2** Data Preparation
    - **1.2.1** Collect datasets
    - **1.2.2** Clean & preprocess data
  - **1.3** Model Development
    - **1.3.1** Configure Med-PaLM environment
    - **1.3.2** Fine-tune model
    - **1.3.3** Evaluate results (accuracy, safety, bias)
  - **1.4** Dashboard / Prototype
    - **1.4.1** Build simple user interface
    - **1.4.2** Integrate model queries & outputs
  - **1.5** Reporting
    - **1.5.1** Draft research report

1.5.2 Finalize APA paper

1.5.3 Create presentation slides

	A	B	C	D	E
2					
3					
4					
5					System Coding (Work Breakdown Structure):
6					**1.0** Med-PaLM Project
7					**1.1** Research & Planning
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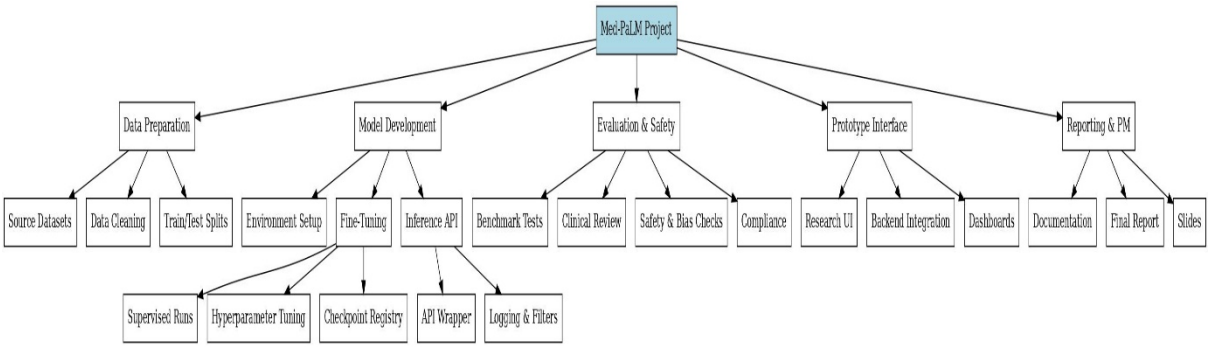
10					**1.2** Data Preparation
11					**1.2.1** Collect datasets
12					**1.2.2** Clean & preprocess data
13					**1.3** Model Development
14					**1.3.1** Configure Med-PaLM environment
15					**1.3.2** Fine-tune model
16					**1.3.3** Evaluate results (accuracy, safety, bias)
17					**1.4** Dashboard / Prototype
18					**1.4.1** Build simple user interface
19					**1.4.2** Integrate model queries & outputs
20					**1.5** Reporting
21					**1.5.1** Draft research report
22					**1.5.2** Finalize APA paper
23					**1.5.3** Create presentation slides
24					



Example Cost Account – 1.3 Model Development

• WP-1.3.1: Install and configure Med-PaLM (Deliverable: working environment).

• WP-1.3.2: Fine-tune on MedQA dataset (Deliverable: tuned model checkpoint).

• WP-1.3.3: Evaluate model using benchmarks (Deliverable: evaluation report with metrics).

• WP-1.3.4: Perform error analysis & bias check (Deliverable: summary of findings and improvements).

Work Package	Deliverable	Estimated Cost (USD)	Responsible Department
WP-1.3.1	Install and configure Med-PaLM (working environment)	$5,000	Machine Learning Engineering (MLE), Compute/Infrastructure Engineer (CIE)
WP-1.3.2	Fine-tune on MedQA dataset (tuned model checkpoint)	$20,000	Machine Learning Engineering (MLE)
WP-1.3.3	Evaluate model using benchmarks (evaluation report with metrics)	$7,500	MLE, Quality Assurance (QA), Clinical SME
WP-1.3.4	Perform error analysis & bias check (summary of findings & improvements)	$10,000	QA, Clinical SME, MLE